

Aditya Nittala

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github.com/AdityaNittala03

Education

Symbiosis Institute of Technology, Nagpur, Bachelor in Computer Science and Engineering Nov 2021 – Apr 2025

- Aggregate GPA: 7.85/10.0

Skills Summary

Languages: Python, C, C++, Go (GoLang), SQL, Dart, HTML, CSS, JavaScript (Vanilla)

Frameworks: Flutter, NoSQL, TailwindCSS, Webpack

Databases: MySQL, PostgreSQL, Redis

AI/ML: NumPy, Pandas, Scikit-learn, TensorFlow, Matplotlib, Seaborn, joblib, PyTorch

Tools: Docker, Git, GitHub, Postman, Android Studio, MySQL, AutoCAD (basic), Unity, Unreal Engine 4

Platforms: Linux, Web, Windows, macOS, Arduino, AWS

IDEs: Visual Studio Code, Google Colab, Xcode, IntelliJ, PyCharm

Abstract Skills: Systems Design (Distributed), Backend Architecture Modeling, Microservices Architecture, Enterprise Architecture, Event Driven Architecture, Caching Strategies (Redis, TTL), Performance Optimization, Latency Reduction, Decoupling Services, CI/CD Pipeline Design

Soft Skills: Leadership, Adaptability, Interpersonal Skills, Event Management, Critical Thinking

Experience

Full-Stack Web Developer Intern, ICEICO Technologies Pvt. Ltd. Jan 2025 – Apr 2025

Project: Migradent Clinic Web Platform

- Identified manual appointment scheduling and paper-based patient records causing 30% administrative inefficiencies and data security risks
- Developed HIPAA-compliant Clinic Management System with online appointment booking and automated WhatsApp notifications, reducing no-shows by 25%
- Built patient record management with encrypted data storage and audit logs, eliminating paper records
- Designed interactive treatment catalog with QA section to improve patient education
- Optimized page load times from 3.8s to 1.6s using Redis caching and WebP image conversion
- Achieved WCAG 2.1 compliance with screen reader support and keyboard navigation
- Implemented rate limiting (60 requests/min) and SQL injection prevention for enhanced security
- Enabled 100+ virtual consultations during testing phase and reduced patient onboarding time by 40%
- Achieved 92/100 Lighthouse performance score
- Tools Used:** Flutter (admin web/mobile app), HTML5, CSS3, JavaScript (ES6+), AWS EC2/S3, Docker

Technical Consultant (Intern), Innovative Hydrocarbons (India) Pvt. Ltd. - Inn Gas Sept 2022 – Nov 2022

Software Module for LPG Station Management

- Identified manual fuel consumption tracking and misreported cash-based transactions by pump operators
- Developed Python module for monitoring pump operators' sales using fingerprint-based tracking and fuel sales across stations with edge devices
- Integrated temperature monitoring capability with factory machines
- Connected module to existing platform for inventory management and station monitoring

- Achieved monthly cost savings of over 2,00,000
- **Tools Used:** Python, Matplotlib, NumPy, Pandas, PySerial, Arduino Board, Optical Fingerprint Sensor

Technical Consultant (Intern), Plastomatic Industries

June 2022 – Aug 2022

Platform-based Inventory Management System

- Developed dashboard system recording inventory consumption and purchases through edge devices (Arduino boards) for consistent data entries and theft prevention
- Implemented temperature monitoring capability integrated with factory machines
- Created automated alerts for low stock levels and input-output ratios
- Designed customer-friendly feedback system increasing survey submissions from 10% to 40%
- Deployed platform on on-premises machines/server
- **Tools Used:** Python, Flutter, Dart, Arduino, Seaborn, Pandas, NumPy, PySerial, Docker

Projects

GoNews: Concurrent News Aggregator

[Github](#)*ink*

- Full-stack Flutter + Go application featuring advanced news aggregation with India-first content strategy
- Backend: Go 1.21 + Fiber framework, PostgreSQL 15+ with full-text search, Redis 7+ caching, JWT authentication
- Advanced Algorithms: 4-layer deduplication engine using Levenshtein distance + SHA256 hashing (<2s performance), dynamic TTL caching (15min-4hr), performance scoring (0-100)
- Features: Multi-dimensional filtering (15+ types), concurrent API integration (NewsData.io, GNews, RapidAPI), IST timezone optimization, market hours detection
- Architecture: Clean architecture with dependency injection, background monitoring, auto-optimization demonstrating enterprise-level system design

AI Finance Assistant

[Github](#)*ink*

- Developed comprehensive AI-powered financial management system using Python Flask backend, PostgreSQL database, and Vanilla JavaScript frontend with D3.js visualizations
- Implemented 4 machine learning models (scikit-learn, TensorFlow) achieving 85%+ accuracy for transaction categorization, spending predictions, and budget recommendations
- Built RESTful API with JWT authentication, Redis caching, and real-time data processing
- Created responsive PWA with interactive charts optimized for Indian Rupees
- Containerized with Docker, featuring continuous learning pipeline and anomaly detection for enhanced financial insights

DuckDB SQL Query Engine using LLM Application

2024

- Built application integrating DuckDB with Large Language Models (LLMs) to enable natural language SQL queries for non-technical users
- Tackled challenges of accurate natural language-to-SQL translation and user input ambiguities using fine-tuned NLP pipeline
- Implemented robust error handling improving query efficiency and reducing analysis time
- Enhanced user experience with simplified data analysis workflow
- **Technologies Used:** DuckDB, Python, OpenAI GPT, NLP, Flask, JavaScript, HTML, CSS

CHIP-8 Emulator - Systems Programming Project

[Github](#)*ink*

- Developed complete CHIP-8 emulator from scratch in C, implementing full CPU architecture with 35 instruction opcodes, 4KB memory management, and 16-register system
- Built real-time graphics rendering using SDL2 with 64x32 pixel display, configurable scaling algorithms, and 60fps frame rate control
- Implemented interactive debugger with breakpoint management, step-through execution, and real-time disassembler using bit manipulation for instruction decoding
- Features include audio synthesis with configurable waveform generation, save/load state functionality using

binary serialization, and cross-platform compatibility

- Utilized mathematical timing calculations for authentic game speed emulation, pointer arithmetic for memory operations, and hexadecimal-to-binary conversion for ROM loading
- **Technologies:** C99, SDL2, CMake/Make build systems, Git version control

Publications

Anemia Detection and Classification from Blood Samples Using Data Analysis and Deep Learning 2024

Aditya Nittala et al.

Automatika Journal (Q2, Taylor and Francis) - Study examining the possibility and impact of utilizing deep learning on blood samples to rapidly and proactively identify underlying health issues. Utilizing effective algorithms to construct models for addressing health problems and determining potential healthcare options based on geographical location.

DOI: 10.1080/00051144.2024.2352317

Optimizing Neural Radiance Field: A Comprehensive Review of the Impact of Different Optimizers 2025

Aditya Nittala et al.

Electrical Engineering International Journal - Neural radiance field (NeRF) models depict 3D scenes from images, producing photorealistic views. This study explores four optimizers—Adam, AdamW, RMSProp, and Adagrad—comparing their effectiveness in reconstruction quality, highlighting optimizer selection's importance.

DOI: 10.11591/eei.v14i1.8315

Anemia Detection and Classification Using Data Analysis of Blood Samples 2023

Aditya Nittala et al.

IEEE Conference - Research exploring the feasibility and impact of using data science over blood samples to quickly and skillfully identify fundamental health problems associated with anemia detection and classification.

DOI: 10.1109/ICAIIHI57871.2023.10489465

Achievements

- Won 1st Prize at "Hack-O-Week 1.2", organized by Google Developer Student Clubs at SIT, Nagpur
- Won 2nd Prize at "A Code Quiz", organized by SRC at SIT, Nagpur, a technical quiz based on programming logic
- Selected to participate in MIT (Massachusetts Institute of Technology) iQuHACK 2023
- Secured 2nd Position at "Skill-O-Mania", Intra-University Men's Basketball, as Captain of the Basketball Team

Extra-Curricular Experience

President, Student's Representative Council 2022 – 2023

- Managed Core Committee (6 members, 7 clubs, 60+ total members) and successfully organized multiple events fostering teamwork, creativity, and effective communication

Workshop Organizer, App Development Using Flutter 2023

- Organized 2-day workshop covering Android Studio and Flutter basics with 14 hours of content
- Achieved 140 registrations with participants creating 3 beautiful apps using Flutter SDK